

3.7.2 Gravitational fields (A-level only)

3.7.2.1 Newton's law (A-level only)

Content	Opportunities for skills development
Gravity as a universal attractive force acting between all matter. Magnitude of force between point masses: $F = \frac{Gm_1m_2}{r^2}$ where G is the gravitational constant.	MS 0.4 Students can estimate the gravitational force between a variety of objects.

3.7.2.2 Gravitational field strength (A-level only)

Content	Opportunities for skills development
Representation of a gravitational field by gravitational field lines. g as force per unit mass as defined by $g = \frac{F}{m}$ Magnitude of g in a radial field given by $g = \frac{GM}{r^2}$	

3.7.2.3 Gravitational potential (A-level only)

Content	Opportunities for skills development
Understanding of definition of gravitational potential, including zero value at infinity. Understanding of gravitational potential difference. Work done in moving mass m given by $\Delta W = m\Delta V$ Equipotential surfaces. Idea that no work is done when moving along an equipotential surface. V in a radial field given by $V = -\frac{GM}{r}$ Significance of the negative sign. Graphical representations of variations of g and V with r. V related to g by: $g = -\frac{\Delta V}{\Delta r}$ ΔV from area under graph of g against r.	MS 3.8, 3.9 Students use graphical representations to investigate relationships between v, r and g.

3.7.2.4 Orbits of planets and satellites (A-level only)

Content	Opportunities for skills development
Orbital period and speed related to radius of circular orbit; derivation of $T^2 \propto r^3$ Energy considerations for an orbiting satellite. Total energy of an orbiting satellite. Escape velocity. Synchronous orbits. Use of satellites in low orbits and geostationary orbits, to include plane and radius of geostationary orbit.	MS 0.4 Estimate various parameters of planetary orbits, eg kinetic energy of a planet in orbit. MS 3.11 Use logarithmic plots to show relationships between T and r for given data.